

ReCycleGuard

Agentic AI for E-Waste EPR Compliance Reconciliation

Turning fragmented compliance records into traceable, evidence-backed reconciliation insights.

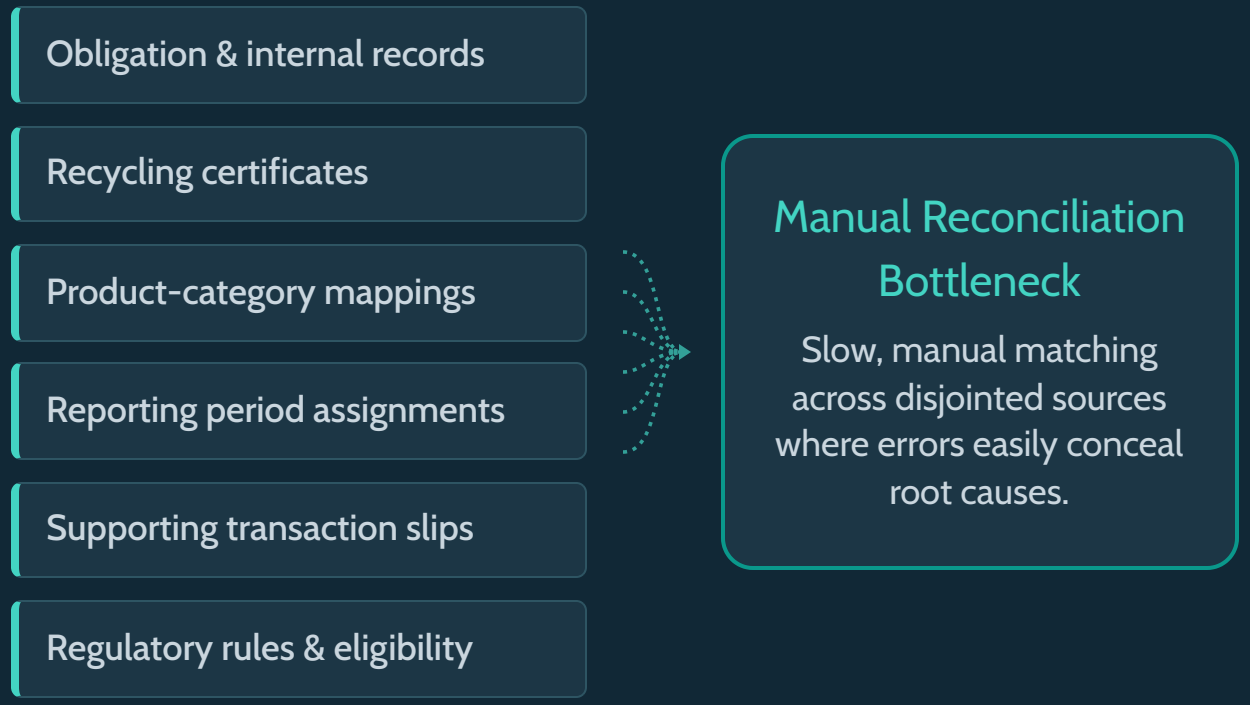
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HYPOTHETICAL WORKFLOW

Scattered records make EPR discrepancies hard to explain

Six disparate record streams feed into a single manual comparison point. That process is slow, and inconsistent documentation hides the underlying cause of mismatches.

No statistical claims are made; represents illustrative workflow friction.



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E-waste and EPR: the context behind the records

E-waste

E-waste: discarded electrical and electronic equipment

EPR

EPR: producers are responsible for the end-of-life of their products

Roles

Producers, recyclers, certificates, and regulatory guidance each play a role

Regulatory context

Context: India's E-Waste (Management) Rules, 2022, amendments, and CPCB guidance

Verified regulation is kept separate from simplified project assumptions

Sources: eprewaste.cpcb.gov.in, cpcb.nic.in, moef.gov.in

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PROPOSED PROTOTYPE ARCHITECTURE

ReCycleGuard is an agentic decision-support system

Objective: to develop an agentic AI prototype that identifies and investigates discrepancies between EPR obligation records and supporting recycling-related records, producing evidence-backed findings for human review.

1

Examine obligation, transaction, and certificate records

2

Flag mismatches and plan the next investigative steps

3

Retrieve regulatory guidance and run verification checks

4

Revisit unresolved issues with more evidence or another check

5

Produce an auditable report: findings, evidence, uncertainty, next actions

It supports compliance officers. It does not declare legal compliance, establish fraud, or file returns. Proposed prototype.

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INVESTIGATION WORKFLOW

The agent investigates in a loop, not a straight line

01	02	03	04	05	06	07	08
Input Records	Initial Analysis	Investigation Plan	Tool-Based Checks	Regulatory Retrieval	Verification Loop	Reconciliation Report	Human Review

ACTIVE FEEDBACK LOOP: Stage 06 (Verification) checks evidence sufficiency. If inconclusive, it routes back to Stage 03 (Plan) rather than guessing.

Compliance Officer Holds Final Decision

Dynamic Planning

Picks specific verification tools per discrepancy rather than executing a rigid static script.

Multi-Tool Toolkit

Executes duplicate certificate checks, unit arithmetic, and period validation.

Refined Retrieval

Queries relevant CPCB regulatory clauses again when initial evidence references are incomplete.

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ILLUSTRATIVE CASE

Example: a 20-unit quantity mismatch

Hypothetical and illustrative, not an official regulatory calculation. The ledger records 100 units against a certificate; the certificate shows 80. Difference: 20 units.

> “Unresolved quantity discrepancy: 20 units. Further verification of allocation records is required.”

A difference alone does not prove non-compliance.

1. Flag the quantity mismatch
2. Check for duplicate certificate or transaction
3. Verify same category and reporting period
4. Retrieve guidance if eligibility or allocation applies
5. Decide whether evidence explains the gap
6. Produce a finding with remaining uncertainty

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RESEARCH BOUNDARIES

A narrow scope keeps the prototype testable

Included

- One hypothetical producer, one reporting period
- A few e-waste product categories
- Structured sample obligation, transaction, and certificate records
- Curated official regulatory documents
- Discrepancy detection, evidence-backed reports, human review

Outside scope

- Every category or regulatory scenario
- Live government portal integration
- Automatic filing of returns
- Legal decisions, certification, or proof of fraud

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SYSTEM OUTPUT

The report shows what is inconsistent, why, and what to check next

Reference & Scope	Discrepancy Detail	Checks & Source Basis	Finding & Next Action
REC-2024-089 Cert #7741 · FY24 Q3	**Quantity Mismatch** Expected: 100 MT Recorded: 80 MT	Duplicate check: Passed Period match: Verified CPCB Notice Cl. 4.2	Pending Verification Unresolved 20 MT variance. Inspect recycler batch credit allocation slips.
REC-2024-104 Cert #8102 · FY24 Q3	**Category Mapping** Target: Cat ITEW1 Found: Cat CEEW2	Schedule table check: Failed Serial audit: Flagged E-Waste Rules 2022 Sch. I	Pending Verification Product code misaligned. Review producer bill-of-materials classification.

Illustrative mock reconciliation report. Status labels and findings are prototype-generated outputs designed to support human compliance officer decision-making.

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Expected impact, to be proven through evaluation

Intended benefits

- Less repetitive manual comparison
- More systematic, traceable investigations
- Uncertainty highlighted, not hidden
- Easier prioritization for reviewers

Evaluation measures

- Detection accuracy on labeled test cases
- Correct arithmetic and record comparisons
- Relevance of retrieved regulatory evidence
- Completeness of evidence references
- Identifying unresolved cases
- Rate of unsupported claims

No improvements are promised before testing.

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ReCycleGuard does not replace compliance expertise

It helps compliance professionals investigate discrepancies systematically, transparently, and with evidence.

Problem: fragmented records and complex discrepancy investigation
Solution: agentic investigation, tools, regulatory retrieval, verification
Outcome: evidence-backed reports that assist human decision-makers

Future directions: more categories, more reporting periods, wider document coverage, integrations. None exist today.

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